MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**AERONAUTICAL ENGINEERING**	**Course Type**	**C**
Course Name	**AVIONICS AND NAVIGATION SYSTEM**	**Course Code**	**AAE 3123**
Semester	**V**	**Max Marks**	**50**
Date	**26/11/2024**	**Duration**	**3hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	How would you interpret the reading shown on the instrument? What does it indicate about the current measurement?  	4	CO1	1,2,3	CLO1	1,2,3,5	3

	 						
1B	Given the message 1101011011 and CRC generator 10011, identify whether the transmission contains an error after applying the Cyclic Redundancy Check (CRC) check.	**4**	**CO2**	**1,5,9**	**CLO2**	**1,2,3**	**3**
1C	A pitot static tube placed in the centre of a 300mm pipe line has one orifice pointing upstream and other perpendicular to it. The mean velocity in the pipe is 0.80 of the central velocity. Find the discharge through the pipe if the pressure difference between the two orifices is 60mm of water. Coefficient of pitot tube is 0.98.	**2**	**CO1**	**1,2,3**	**CLO1**	**1,2,3,5**	**3**
2A	Elucidate the significance of **Threshold Detction** by deriving the mathematical expressions for **RADAR Range calculation by considering ambiguous and unambiguous** measurement.	**4**	**CO2**	**1,5,9**	**CLO2**	**1,2,3**	**2**
2B	A frequency modulated voltage wave is given by the equation, $E=12\cos(6\times10^8 t+5 \sin 1250t)$. Find (i) carrier frequency (ii) signal frequency (iii) modulation index (iv) maximum frequency deviation (v) power dissipated by the FM wave in a 10 ohm resistor.	**3**	**CO4**	**1,2,3**	**CLO4**	**3,4,9,12**	**3**
2C	Simplify the Boolean expression F(A, B, C)=∑(1,5,2,4,7) using Kmap.	**3**	**CO2**	**1,5,9**	**CLO2**	**1,2,3**	**3**
3A	An observer at the geographical north pole has a GPS receiver. At an instant in time, four GPS satellites all have the same range from the observer, and the GPS receiver records a measured delay time for the C/A signal of 0.17097528 s for each satellite. The four satellites' coordinates are calculated to be (0, -13280.5, 23002.5), (0, 13280.5, 23002.5), (-13280.5, 0, 23002.5), (13280.5, 0, 23002.5), where all distances are in km. Assuming an earth radius of 6378.0 km at the north pole, so that the observer's coordinates are (0,0, 6378), **determine the clock offset error in the GPS receiver.** (velocity of light in free space to be 2.99792458×10^8 m/s.)	**4**	**CO2**	**1,5,9**	**CLO4**	**3,4,9,12**	**3**

<table>
<tr><td>3B</td><td>Explain the fundamental instructions required for executing the addition of two numbers using an 8085 Microprocessor with an example.</td><td>3</td><td>CO6</td><td>3,5,9,12</td><td>CLO6</td><td>4,5,9,12</td><td>2</td></tr>
<tr><td>3C</td><td>During a flight, an aircraft's radar system begins displaying incorrect readings, leading to misinterpretation of weather conditions and potential obstacles. This results in the flight crew making unnecessary changes to the flight path. The altimeter also shows incorrect altitude readings for brief periods, complicating decision-making, particularly during approaches in poor visibility. Fortunately, the issue is detected early, and the crew switches to manual radar operations, preventing a more serious risk.
1. Discuss how electromagnetic interference (EMI) from external sources can affect avionics systems and lead to these types of failures.
2. How can avionics systems be designed to prevent such malfunctions and ensure reliable performance during flight?</td><td>3</td><td>CO5</td><td>1,2</td><td>CLO5</td><td>2,3, 9,12</td><td>4</td></tr>
<tr><td>4A</td><td>Explain how an Inertial Measurement System (IMU) determines orientation and movement of an aircraft.</td><td>3</td><td>CO4</td><td>1,2,3</td><td>CLO4</td><td>3,4,9,12</td><td>2</td></tr>
<tr><td>4B</td><td>Design a PID controller using Ziegler-Nichols second method for the given open loop system
$$G(S) = \frac{1}{S(S+1)(S+5)}$$
Mention one potential safety consideration associated with the integration of intelligent systems which can enhance the performance of an Automatic Flight Control System (AFCS).</td><td>4</td><td>CO3</td><td>1,2,3,5, 9</td><td>CLO3</td><td>1,2,3,4</td><td>4</td></tr>
<tr><td>4C</td><td>Investigate the different communication bus systems used in modern aircraft and evaluate their roles in ensuring efficient and reliable data transmission.</td><td>3</td><td>CO5</td><td>1,2</td><td>CLO5</td><td>2,3, 9,12</td><td>3</td></tr>
<tr><td>5A</td><td>If a UAV operator wants to operate in a restricted airspace, what steps must they take to comply with DGCA regulations and what avionics software is necessary for managing and controlling the electronic systems onboard.</td><td>3</td><td>CO6</td><td>3,5,9,12</td><td>CLO6</td><td>4,5,9,12</td><td>3</td></tr>
<tr><td>5B</td><td>With the help of Daigram, explain the Standard CAN arbitration process, comment on which ID/Node going to win the bus access and why.
<table><tr><td>CAN NODE</td><td>Identifier (HEX)</td><td>Identifier (Binary)</td></tr><tr><td>1</td><td>0×7F3</td><td>11111110011</td></tr><tr><td>2</td><td>0×6B3</td><td>11010110011</td></tr></table></td><td>4</td><td>CO5</td><td>1,2</td><td>CLO5</td><td>2,3, 9,12</td><td>3</td></tr>
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	3	0×6D9	11011011001						
5C	Explain the role of each component in an aircraft's power generation system and describe how power is distributed to various subsystems during flight operations.			**3**	**CO4**	**1,2,3**	**CLO4**	**3,4,9,12**	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***6***
2	***15***
3	***4***
4	***9***
5	***10***
6	***6***

BL	Total Marks
2	**13**
3	**30**
4	**7**
5	